

How the recloser operating times and the fuse melting times are calculated

Replication of Yousaf et al. (2022), IEEE 13-node feeder, Table IV. All numbers are taken from the PowerFactory replication in *Replication/results* (studies.json, settings.json, Table_IV.csv).

1. Summary

Each operating time is obtained in three steps: **(a)** the fault current through the recloser is read from the PowerFactory short-circuit study, **(b)** it is divided by the pickup setting to get the multiple M , and **(c)** M is put into the relay's time-current curve and scaled by the time dial. The fast and the delayed operation use the same curve with two different dials (and, for R2, two different plug settings). The fuse time t_{MMT} of Table IV is read from the fuse's minimum-melting curve at the current through the fuse (section 4.3).

The two reclosers use different curves, so there are two calculation methods: R1 (GE IAC77B801A) has a closed formula, R2 (GE/Alstom CDG34) has a manufacturer's table that is interpolated. The replicated times do **not** equal the paper's Table IV: R1 is 12 % to 78 % slower in eleven of twelve rows (bus 652 is 34 % faster). The R1 formula itself is confirmed by the paper's own worked example (section 3), so the difference comes from the fault currents; for R2 the paper does not publish the settings, so they had to be assumed (section 6).

2. Which operation each column of Table IV is

The paper's eqs. (1) and (2) give the **forward** fast and delayed times, eqs. (10) and (11) the **reverse** ones. R1 only ever sees current from the grid, so both R1 columns are forward for every bus. R2 is the dual-setting directional recloser: it operates forward for faults downstream of it and reverse (DG current flowing back towards 632) for faults upstream of it.

Faulted bus	R1 fast / delayed	R2 fast / delayed	Formula in the paper
632, 633, 645, 646, DL (upstream of R2)	forward	reverse setting group	R1: eqs. (1), (2); R2: eqs. (10), (11)
671, 692, 675, 680, 684, 611, 652 (downstream of R2)	forward	forward setting group	R1 and R2: eqs. (1), (2)

3. Settings used in the calculation

Pickup, eq. (3) of the paper (eq. (12) for the reverse direction), with OLF = 1.25:

$$I_p = \text{OLF} \times I_{\text{nom}}$$

Recloser	Load current I_{nom}	$1.25 \times I_{\text{nom}}$	Setting actually used	Time dial fast / delayed
R1 IAC77B801A, forward	587.7 A (load flow, DG out)	734.6 A	tap 4 A on CT 900/5 = 720 A	TDS 0.5 / 10
R2 CDG34, forward	470.0 A (load flow, DG out)	587.5 A	CT 1000/5: fast plug 1.5 A = 300 A , delayed plug 3.0 A = 600 A	TMS 0.1 / 1.0
R2 CDG34, reverse	252.2 A (load flow, DG in, towards 632)	315.2 A	CT 500/5: fast plug 1.5 A = 150 A , delayed plug 3.0 A = 300 A	TMS 0.1 / 1.0

For R2 the fast plug is half of $1.25 \times I_{\text{nom}}$ and the delayed plug equals $1.25 \times I_{\text{nom}}$, because the CDG curve only begins at $2 \times$ the plug setting. The fast curve therefore starts at about I_p (600 A forward) and the delayed curve at $2 I_p$ (1200 A forward). Only the R1 time dials are printed in the paper; the basis of every other value is given in section 6.

4. The two calculation methods

4.1 R1: GE IAC extremely inverse equation

The paper writes the recloser time in the general form of eqs. (1) and (2):

$$t = TDS \left[\frac{A}{(I_f/I_p)^n - 1} + B \right]$$

but does not give A, B and n. For the IAC77B801A the model uses the manufacturer's extremely inverse equation (PowerFactory library type *IAC Extremely Inverse GES7005B*), valid from 1.5 to 40 times pickup:

$$t = TDS \left[0.004 + \frac{0.6379}{M - 0.62} + \frac{1.7872}{(M - 0.62)^2} + \frac{0.2461}{(M - 0.62)^3} \right], \quad M = \frac{I_f}{I_p}$$

Check against the paper. The paper gives one worked example: for 4219 A downstream of F645, R1 operates in 0.097 s (fast) and 1.932 s (delayed). The same current in the equation above:

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 4219.0 / 720$	5.8597
Shifted multiple	$x = M - 0.62 = 5.8597 - 0.62$	5.2397
First term	$0.6379 / x = 0.6379 / 5.2397$	0.12174
Second term	$1.7872 / x^2 = 1.7872 / 27.4547$	0.06510
Third term	$0.2461 / x^3 = 0.2461 / 143.855$	0.00171
Curve value for TDS = 1	$K = 0.004 + 0.12174 + 0.06510 + 0.00171$	0.19255 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.19255$	0.096 s
Delayed (TDS = 10)	$t_D = 10 \times 0.19255$	1.926 s

Result 0.096 s and 1.926 s against the paper's 0.097 s and 1.932 s (-0.7 % and -0.3 %). With the standard IEEE extremely inverse constants in eq. (1) ($A = 28.2$, $B = 0.1217$, $n = 2$) the same current would give 0.484 s and 9.68 s, five times too slow, so the IAC equation is the one the paper used. In every row of the paper's Table IV the R1 delayed time is 20 times the fast time, which is the ratio of the two dials, 10 / 0.5.

4.2 R2: GE/Alstom CDG34, extremely inverse table

The CDG extremely inverse characteristic (PowerFactory library type *Extremely Inverse No. 398.S23.37 CDG14*) is not a formula but a table of time against $M = I_f / I_s$, with one column per time multiplier from 0.1 to 1.0. The settings used here are exactly the first column (TMS 0.1, fast) and the last column (TMS 1.0, delayed), so only M has to be interpolated. Between two table rows the curve is taken as a straight line on log-log axes:

$$t = t_1 \left(\frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(M/M_1)}{\ln(M_2/M_1)}, \quad M = \frac{I_f}{I_s}$$

Part of the table (seconds). Below $M = 2$ the relay does not operate; above $M = 40$ the time stays at the $M = 40$ value.

$M = I_f / I_s$	2	3	4	5	6	7	8	9	10	11	12
TMS 0.1 (fast)	1.155	0.4311	0.2385	0.16	0.1189	0.094	0.07779	0.0672	0.06	0.05488	0.05116
TMS 1.0 (delayed)	17.2	6.218	3.168	2	1.435	1.08	0.831	0.6757	0.5802	0.517	0.472

4.3 Fuses: minimum melting time t_{MMT} from the fuse curve

The paper describes a fuse by the straight line of eq. (6), $\log t = a \log I + b$ with $a = -1.8$, and uses it to find the size that is needed (Table III). The time printed in Table IV is the melting time of the fuse that is actually installed. A fuse has no formula: the manufacturer gives the **minimum-melting** curve (the fuse element starts to melt, t_{MMT}) and the **total-clearing** curve (the arc is out, t_{TCT}) as points. The model uses the PowerFactory library fuses Gould-Shawmut A055C (5.5 kV, E-rated) and reads the curve at the fuse current with the same log-log interpolation as for the CDG table:

$$t_{MMT} = t_1 \left(\frac{t_2}{t_1} \right)^f, \quad f = \frac{\ln(I_{fuse}/I_1)}{\ln(I_2/I_1)}$$

I_{fuse} is the current through the fuse, not the recloser current. With the DG in service a fuse between the two sources and the fault carries the grid share and the DG share together, so it sees more current than either recloser. Example at bus 633: R1 carries 3970 A, R2 (reverse) 1500 A, the fuse F633 5426 A.

Minimum-melting points of A055C400E (F633), the fuse of the first example:

Current (A)	6692	7049	7483	8025	8728	9688	11108	13516	19000
t_{MMT} (s)	0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01

Coordination needs t_{MMT} to be longer than the recloser's fast trip (fuse saving) and t_{TCT} to be shorter than the delayed trip (the fuse clears a permanent fault before the recloser locks out).

5. Worked examples for five buses

Fault currents are from the PowerFactory short-circuit study with the DG in service (4.05 MVA at 692) and a bolted fault. The current used for each recloser is the largest of its three phase currents. Buses 633 and 646 are in R1's zone (R2 operates in reverse); buses 671, 675, 611 are downstream of R2 (R2 operates forward).

Example 1: bus 633, three-phase (LLL) fault on phases ABC

Current through R1: **3970.2 A**. Current through R2: **1499.7 A** (reverse, DG contribution only).

R1 forward, $I_p = 720$ A

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 3970.2 / 720$	5.5141
Shifted multiple	$x = M - 0.62 = 5.5141 - 0.62$	4.8941
First term	$0.6379 / x = 0.6379 / 4.8941$	0.13034
Second term	$1.7872 / x^2 = 1.7872 / 23.9522$	0.07462
Third term	$0.2461 / x^3 = 0.2461 / 117.224$	0.00210
Curve value for TDS = 1	$K = 0.004 + 0.13034 + 0.07462 + 0.00210$	0.21106 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.21106$	0.106 s
Delayed (TDS = 10)	$t_D = 10 \times 0.21106$	2.111 s

R2 reverse, fast: $I_s = 150$ A, TMS 0.1

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1499.7 / 150$	9.9983
Table rows around M (TMS 0.1)	$M_1 = 9 \rightarrow t_1 = 0.0672$ s; $M_2 = 10 \rightarrow t_2 = 0.06$ s	
Position between the rows	$f = \ln(9.9983 / 9) / \ln(10 / 9) = 0.10519 / 0.10536$	0.9984
Fast (TMS = 0.1)	$t = t_1 \times (t_2 / t_1)^f = 0.0672 \times (0.89286)^{0.9984}$	0.060 s

R2 reverse, delayed: $I_s = 300$ A, TMS 1.0

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1499.7 / 300$	4.9991
Table rows around M (TMS 1.0)	$M_1 = 4 \rightarrow t_1 = 3.168$ s; $M_2 = 5 \rightarrow t_2 = 2$ s	
Position between the rows	$f = \ln(4.9991 / 4) / \ln(5 / 4) = 0.22297 / 0.22314$	0.9992
Delayed (TMS = 1.0)	$t = t_1 \times (t_2 / t_1)^f = 3.168 \times (0.63131)^{0.9992}$	2.001 s

Fuse F633 (A055C400E): current through the fuse 5425.9 A, minimum-melting curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 4800 \text{ A} \rightarrow t_1 = 0.2 \text{ s}; I_2 = 6392 \text{ A} \rightarrow t_2 = 0.1 \text{ s}$	
Position between the points	$f = \ln(5425.9 / 4800) / \ln(6392 / 4800) = 0.12256 / 0.28643$	0.4279
t_{MMT}	$t = t_1 \times (t_2 / t_1)^f = 0.2 \times (0.50000)^{0.4279}$	0.149 s

Fuse F633, total-clearing curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 4901 \text{ A} \rightarrow t_1 = 0.3 \text{ s}; I_2 = 5800 \text{ A} \rightarrow t_2 = 0.2 \text{ s}$	
Position between the points	$f = \ln(5425.9 / 4901) / \ln(5800 / 4901) = 0.10174 / 0.16842$	0.6041
t_{TCT}	$t = t_1 \times (t_2 / t_1)^f = 0.3 \times (0.66667)^{0.6041}$	0.235 s

Fuse saving: $t_{\text{MMT}} 0.149 \text{ s} - \text{last fast trip } 0.106 \text{ s} = \mathbf{+0.043 \text{ s}}$ (the recloser trips first). Permanent fault: $t_{\text{TCT}} 0.235 \text{ s}$, first delayed trip 2.001 s (the fuse clears first).

Bus 633	R1 fast	R1 delayed	R2 fast (rev)	R2 delayed (rev)	Fuse t_{MMT}
Calculated above	0.106 s	2.111 s	0.060 s	2.001 s	0.149 s
Paper, Table IV	0.094 s	1.880 s	0.052 s	0.706 s	0.163 s
Difference	+12 %	+12 %	+15 %	+183 %	-9 %

Example 2: bus 646, line-to-line (LL) fault on phases BC

Current through R1: **2784.8 A**. Current through R2: **1114.0 A** (reverse, DG contribution only).

R1 forward, $I_p = 720 \text{ A}$

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 2784.8 / 720$	3.8678
Shifted multiple	$x = M - 0.62 = 3.8678 - 0.62$	3.2478
First term	$0.6379 / x = 0.6379 / 3.2478$	0.19641
Second term	$1.7872 / x^2 = 1.7872 / 10.5479$	0.16944
Third term	$0.2461 / x^3 = 0.2461 / 34.257$	0.00718
Curve value for TDS = 1	$K = 0.004 + 0.19641 + 0.16944 + 0.00718$	0.37703 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.37703$	0.189 s
Delayed (TDS = 10)	$t_D = 10 \times 0.37703$	3.770 s

R2 reverse, fast: $I_s = 150 \text{ A}$, TMS 0.1

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1114.0 / 150$	7.4270
Table rows around M (TMS 0.1)	$M_1 = 7 \rightarrow t_1 = 0.094 \text{ s}; M_2 = 8 \rightarrow t_2 = 0.07779 \text{ s}$	
Position between the rows	$f = \ln(7.4270 / 7) / \ln(8 / 7) = 0.05921 / 0.13353$	0.4434
Fast (TMS = 0.1)	$t = t_1 \times (t_2 / t_1)^f = 0.094 \times (0.82755)^{0.4434}$	0.086 s

R2 reverse, delayed: $I_s = 300 \text{ A}$, TMS 1.0

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1114.0 / 300$	3.7135
Table rows around M (TMS 1.0)	$M_1 = 3 \rightarrow t_1 = 6.218 \text{ s}; M_2 = 4 \rightarrow t_2 = 3.168 \text{ s}$	
Position between the rows	$f = \ln(3.7135 / 3) / \ln(4 / 3) = 0.21336 / 0.28768$	0.7417
Delayed (TMS = 1.0)	$t = t_1 \times (t_2 / t_1)^f = 6.218 \times (0.50949)^{0.7417}$	3.771 s

Fuse F646 (A055C400E): current through the fuse 3671.9 A, minimum-melting curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 3446 \text{ A} \rightarrow t_1 = 0.5 \text{ s}; I_2 = 3718 \text{ A} \rightarrow t_2 = 0.4 \text{ s}$	
Position between the points	$f = \ln(3671.9 / 3446) / \ln(3718 / 3446) = 0.06350 / 0.07597$	0.8358
t_{MMT}	$t = t_1 \times (t_2 / t_1)^f = 0.5 \times (0.80000)^{0.8358}$	0.415 s

Fuse F646, total-clearing curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 3645 \text{ A} \rightarrow t_1 = 0.7 \text{ s}; I_2 = 3827 \text{ A} \rightarrow t_2 = 0.6 \text{ s}$	
Position between the points	$f = \ln(3671.9 / 3645) / \ln(3827 / 3645) = 0.00736 / 0.04872$	0.1510
t_{TCT}	$t = t_1 \times (t_2 / t_1)^f = 0.7 \times (0.85714)^{0.1510}$	0.684 s

Fuse saving: $t_{\text{MMT}} 0.415 \text{ s} - \text{last fast trip } 0.189 \text{ s} = \mathbf{+0.226 \text{ s}}$ (the recloser trips first). Permanent fault: $t_{\text{TCT}} 0.684 \text{ s}$, first delayed trip 3.770 s (the fuse clears first).

Bus 646	R1 fast	R1 delayed	R2 fast (rev)	R2 delayed (rev)	Fuse t_{MMT}
Calculated above	0.189 s	3.770 s	0.086 s	3.771 s	0.415 s
Paper, Table IV	0.128 s	2.555 s	0.103 s	2.046 s	0.181 s
Difference	+47 %	+48 %	-16 %	+84 %	+129 %

Example 3: bus 671, three-phase (LLL) fault on phases ABC

Current through R1: **3412.3 A**. Current through R2: **3391.0 A** (forward, from the grid).

R1 forward, $I_p = 720$ A

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 3412.3 / 720$	4.7393
Shifted multiple	$x = M - 0.62 = 4.7393 - 0.62$	4.1193
First term	$0.6379 / x = 0.6379 / 4.1193$	0.15486
Second term	$1.7872 / x^2 = 1.7872 / 16.9682$	0.10533
Third term	$0.2461 / x^3 = 0.2461 / 69.896$	0.00352
Curve value for TDS = 1	$K = 0.004 + 0.15486 + 0.10533 + 0.00352$	0.26771 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.26771$	0.134 s
Delayed (TDS = 10)	$t_D = 10 \times 0.26771$	2.677 s

R2 forward, fast: $I_s = 300$ A, TMS 0.1

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 3391.0 / 300$	11.3033
Table rows around M (TMS 0.1)	$M_1 = 11 \rightarrow t_1 = 0.05488$ s; $M_2 = 12 \rightarrow t_2 = 0.05116$ s	
Position between the rows	$f = \ln(11.3033 / 11) / \ln(12 / 11) = 0.02720 / 0.08701$	0.3126
Fast (TMS = 0.1)	$t = t_1 \times (t_2 / t_1)^f = 0.05488 \times (0.93222)^{0.3126}$	0.054 s

R2 forward, delayed: $I_s = 600$ A, TMS 1.0

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 3391.0 / 600$	5.6517
Table rows around M (TMS 1.0)	$M_1 = 5 \rightarrow t_1 = 2$ s; $M_2 = 6 \rightarrow t_2 = 1.435$ s	
Position between the rows	$f = \ln(5.6517 / 5) / \ln(6 / 5) = 0.12251 / 0.18232$	0.6720
Delayed (TMS = 1.0)	$t = t_1 \times (t_2 / t_1)^f = 2 \times (0.71750)^{0.6720}$	1.600 s

Bus 671	R1 fast	R1 delayed	R2 fast (fwd)	R2 delayed (fwd)	Fuse t_{MMT}
Calculated above	0.134 s	2.677 s	0.054 s	1.600 s	no fuse
Paper, Table IV	0.105 s	2.091 s	0.068 s	1.113 s	---
Difference	+27 %	+28 %	-21 %	+44 %	---

Example 4: bus 675, three-phase (LLL) fault on phases ABC

Current through R1: **3090.3 A**. Current through R2: **3049.6 A** (forward, from the grid).

R1 forward, $I_p = 720$ A

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 3090.3 / 720$	4.2921
Shifted multiple	$x = M - 0.62 = 4.2921 - 0.62$	3.6721
First term	$0.6379 / x = 0.6379 / 3.6721$	0.17372
Second term	$1.7872 / x^2 = 1.7872 / 13.4843$	0.13254
Third term	$0.2461 / x^3 = 0.2461 / 49.516$	0.00497
Curve value for TDS = 1	$K = 0.004 + 0.17372 + 0.13254 + 0.00497$	0.31522 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.31522$	0.158 s
Delayed (TDS = 10)	$t_D = 10 \times 0.31522$	3.152 s

R2 forward, fast: $I_s = 300$ A, TMS 0.1

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 3049.6 / 300$	10.1654
Table rows around M (TMS 0.1)	$M_1 = 10 \rightarrow t_1 = 0.06$ s; $M_2 = 11 \rightarrow t_2 = 0.05488$ s	
Position between the rows	$f = \ln(10.1654 / 10) / \ln(11 / 10) = 0.01640 / 0.09531$	0.1721
Fast (TMS = 0.1)	$t = t_1 \times (t_2 / t_1)^f = 0.06 \times (0.91467)^{0.1721}$	0.059 s

R2 forward, delayed: $I_s = 600$ A, TMS 1.0

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 3049.6 / 600$	5.0827
Table rows around M (TMS 1.0)	$M_1 = 5 \rightarrow t_1 = 2$ s; $M_2 = 6 \rightarrow t_2 = 1.435$ s	
Position between the rows	$f = \ln(5.0827 / 5) / \ln(6 / 5) = 0.01640 / 0.18232$	0.0900
Delayed (TMS = 1.0)	$t = t_1 \times (t_2 / t_1)^f = 2 \times (0.71750)^{0.0900}$	1.941 s

Fuse F692-R (A055C250E): current through the fuse 5022.1 A, minimum-melting curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 4848$ A $\rightarrow t_1 = 0.07$ s; $I_2 = 5203$ A $\rightarrow t_2 = 0.06$ s	
Position between the points	$f = \ln(5022.1 / 4848) / \ln(5203 / 4848) = 0.03527 / 0.07067$	0.4991
t_{MMT}	$t = t_1 \times (t_2 / t_1)^f = 0.07 \times (0.85714)^{0.4991}$	0.065 s

Fuse F692-R, total-clearing curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 3700$ A $\rightarrow t_1 = 0.2$ s; $I_2 = 5170$ A $\rightarrow t_2 = 0.1$ s	
Position between the points	$f = \ln(5022.1 / 3700) / \ln(5170 / 3700) = 0.30551 / 0.33454$	0.9132
t_{TCT}	$t = t_1 \times (t_2 / t_1)^f = 0.2 \times (0.50000)^{0.9132}$	0.106 s

Fuse saving: $t_{MMT} - 0.065$ s – last fast trip 0.059 s = **+0.006 s** (the recloser trips first). Permanent fault: $t_{TCT} - 0.106$ s, first delayed trip 1.941 s (the fuse clears first).

Bus 675	R1 fast	R1 delayed	R2 fast (fwd)	R2 delayed (fwd)	Fuse t_{MMT}
Calculated above	0.158 s	3.152 s	0.059 s	1.941 s	0.065 s
Paper, Table IV	0.121 s	2.414 s	0.074 s	1.332 s	0.185 s
Difference	+30 %	+31 %	-20 %	+46 %	-65 %

Example 5: bus 611, single line-to-ground (LG) fault on phases C

Current through R1: **1814.2 A**. Current through R2: **1718.3 A** (forward, from the grid).

R1 forward, $I_p = 720$ A

Step	Working	Result
Multiple of pickup	$M = I_f / I_p = 1814.2 / 720$	2.5197
Shifted multiple	$x = M - 0.62 = 2.5197 - 0.62$	1.8997
First term	$0.6379 / x = 0.6379 / 1.8997$	0.33580
Second term	$1.7872 / x^2 = 1.7872 / 3.6087$	0.49524
Third term	$0.2461 / x^3 = 0.2461 / 6.855$	0.03590
Curve value for TDS = 1	$K = 0.004 + 0.33580 + 0.49524 + 0.03590$	0.87094 s
Fast (TDS = 0.5)	$t_F = 0.5 \times 0.87094$	0.435 s
Delayed (TDS = 10)	$t_D = 10 \times 0.87094$	8.709 s

R2 forward, fast: $I_s = 300$ A, TMS 0.1

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1718.3 / 300$	5.7278
Table rows around M (TMS 0.1)	$M_1 = 5 \rightarrow t_1 = 0.16$ s; $M_2 = 6 \rightarrow t_2 = 0.1189$ s	
Position between the rows	$f = \ln(5.7278 / 5) / \ln(6 / 5) = 0.13590 / 0.18232$	0.7454
Fast (TMS = 0.1)	$t = t_1 \times (t_2 / t_1)^f = 0.16 \times (0.74313)^{0.7454}$	0.128 s

R2 forward, delayed: $I_s = 600$ A, TMS 1.0

Step	Working	Result
Multiple of plug setting	$M = I_f / I_s = 1718.3 / 600$	2.8639
Table rows around M (TMS 1.0)	$M_1 = 2 \rightarrow t_1 = 17.2$ s; $M_2 = 3 \rightarrow t_2 = 6.218$ s	
Position between the rows	$f = \ln(2.8639 / 2) / \ln(3 / 2) = 0.35904 / 0.40547$	0.8855
Delayed (TMS = 1.0)	$t = t_1 \times (t_2 / t_1)^f = 17.2 \times (0.36151)^{0.8855}$	6.986 s

Fuse F684 (A055C200E): current through the fuse 2235.8 A, minimum-melting curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 2155 \text{ A} \rightarrow t_1 = 0.3 \text{ s}; I_2 = 2500 \text{ A} \rightarrow t_2 = 0.2 \text{ s}$	
Position between the points	$f = \ln(2235.8 / 2155) / \ln(2500 / 2155) = 0.03682 / 0.14850$	0.2480
t_{MMT}	$t = t_1 \times (t_2 / t_1)^f = 0.3 \times (0.66667)^{0.2480}$	0.271 s

Fuse F684, total-clearing curve

Step	Working	Result
Table points around I_{fuse}	$I_1 = 2088 \text{ A} \rightarrow t_1 = 0.5 \text{ s}; I_2 = 2263 \text{ A} \rightarrow t_2 = 0.4 \text{ s}$	
Position between the points	$f = \ln(2235.8 / 2088) / \ln(2263 / 2088) = 0.06841 / 0.08048$	0.8499
t_{TCT}	$t = t_1 \times (t_2 / t_1)^f = 0.5 \times (0.80000)^{0.8499}$	0.414 s

Fuse saving: t_{MMT} 0.271 s – last fast trip 0.128 s = **+0.143 s** (the recloser trips first). Permanent fault: t_{TCT} 0.414 s, first delayed trip 6.986 s (the fuse clears first).

Bus 611	R1 fast	R1 delayed	R2 fast (fwd)	R2 delayed (fwd)	Fuse t_{MMT}
Calculated above	0.435 s	8.709 s	0.128 s	6.986 s	0.271 s
Paper, Table IV	0.326 s	6.523 s	0.146 s	4.385 s	0.417 s
Difference	+34 %	+34 %	-12 %	+59 %	-35 %

All twelve rows of Table IV

The same two calculations for every bus (model value / paper value, seconds). The examples of section 5 are highlighted. The last column is the current R1 would need in order to give the paper's delayed time with the settings of section 3.

Bus	Fault	I_{R1} (A)	R1 fast	R1 delayed	R2	I_{R2} (A)	R2 fast	R2 delayed	I_{R1} from paper (A)
632	LLL	4734	0.081 / 0.070	1.627 / 1.398	rev	1813	0.051 / 0.045	1.416 / 0.514	5271
633	LLL	3970	0.106 / 0.094	2.111 / 1.880	rev	1500	0.060 / 0.052	2.001 / 0.706	4287
645	LL	3124	0.155 / 0.105	3.095 / 2.095	rev	1232	0.075 / 0.087	3.002 / 1.659	3989
646	LL	2785	0.189 / 0.128	3.770 / 2.555	rev	1114	0.086 / 0.103	3.771 / 2.046	3513
DL	LLL	3954	0.106 / 0.087	2.123 / 1.733	rev	1972	0.048 / 0.076	1.213 / 1.402	4531
671	LLL	3412	0.134 / 0.105	2.677 / 2.091	fwd	3391	0.054 / 0.068	1.600 / 1.113	3994
692	LLL	3412	0.134 / 0.115	2.677 / 2.307	fwd	3391	0.054 / 0.072	1.600 / 1.258	3748
675	LLL	3090	0.158 / 0.121	3.152 / 2.414	fwd	3050	0.059 / 0.074	1.941 / 1.332	3641
684	LL	2543	0.222 / 0.125	4.437 / 2.507	fwd	2543	0.072 / 0.075	2.813 / 1.394	3555
680	LLL	2789	0.188 / 0.137	3.761 / 2.747	fwd	2757	0.066 / 0.079	2.381 / 1.555	3358
611	LG	1814	0.435 / 0.326	8.709 / 6.523	fwd	1718	0.128 / 0.146	6.986 / 4.385	2082
652	LG	1795	0.446 / 0.676	8.919 / 13.525	fwd	1761	0.123 / 0.143	6.570 / 9.370	1500

Fuse column of Table IV

Fuse nearest to the fault on the path from the reclosers, its size after step 9 of the method, the current through it and the two fuse times. Margin = $t_{MMT} - \text{last fast trip (R2 for faults downstream of R2; the later of R1 and R2 reverse upstream of it)}$. Buses 632, 671 and 680 are on the main feeder and have no fuse.

Bus	Fault	Fuse	Size	I_{fuse} (A)	t_{MMT} model / paper	Diff.	t_{TCT} (s)	Last fast trip (s)	Margin (s)	Status
633	LLL	F633	400E	5426	0.149 / 0.163	-9 %	0.235	0.106	+0.043	held
645	LL	F632	500E	4191	0.613 / 0.276	+122 %	0.970	0.155	+0.458	held
646	LL	F646	400E	3672	0.415 / 0.181	+129 %	0.684	0.189	+0.226	held
DL	LLL	F-DL	400E	5902	0.121 / 0.248	-51 %	0.193	0.106	+0.015	held
692	LLL	F671-1	400E	3391	0.525 / 0.175	+200 %	0.891	0.054	+0.471	held
675	LLL	F692-R	250E	5022	0.065 / 0.185	-65 %	0.106	0.059	+0.006	held
684	LL	F671-2	300E	3963	0.148 / 0.205	-28 %	0.230	0.072	+0.076	held
611	LG	F684	200E	2236	0.271 / 0.417	-35 %	0.414	0.128	+0.143	held
652	LG	F652	150E	2239	0.138 / 0.233	-41 %	0.193	0.123	+0.015	held

6. Assumptions and their basis

The paper states the relay types, OLF = 1.25 and R1's two time dials. Everything else below had to be chosen; the right-hand column says on what basis.

#	Value used	In the paper?	Basis
1	R1 time dials 0.5 (fast) and 10 (delayed)	Yes	Section IV-A of the paper.

#	Value used	In the paper?	Basis
2	R1 curve is the GE IAC extremely inverse equation, not eq. (1) with IEEE or IEC constants	Relay type only	The paper names the IAC77B801A but gives no A, B, n. This equation reproduces the paper's worked example ($4219 \text{ A} \rightarrow 0.097 / 1.932 \text{ s}$) as $0.096 / 1.926 \text{ s}$; IEEE constants give $0.484 / 9.68 \text{ s}$.
3	R1 pickup 720 A (tap 4 A on a $900/5 \text{ CT}$) instead of $1.25 \times 587.7 = 734.6 \text{ A}$	No	A relay is set on discrete taps. The CT ratio is my choice. Solving the paper's worked example backwards for the pickup gives 722 A , so 720 A fits better than 734.6 A (which gives $0.099 / 1.984 \text{ s}$).
4	R2 curve is the CDG14 extremely inverse table, interpolated log-log between rows	Relay type only	The paper names the CDG34 with extremely inverse characteristic. PowerFactory's own relay and fuse models agree with this calculation on 1896 operating times, largest deviation 1.22 %.
5	R2 forward plug settings 300 A (fast) and 600 A (delayed) on a $1000/5 \text{ CT}$	No	$1.25 \times 470.0 \text{ A} = 587.5 \text{ A}$ by eq. (3). The CDG curve begins at $2 \times$ plug, and in the paper's Figs. 8 and 13 R2's fast curve begins near I_p and the delayed curve near $2 I_p$. This is the largest assumption: the paper never prints R2's settings.
6	R2 time multipliers 0.1 (fast) and 1.0 (delayed)	No	Fast: the lowest dial of the relay. Delayed: the highest dial that still operates at least 10 cycles before R1's delayed curve for every fault downstream of R2 (the paper's criterion for the two reclosers).
7	R2 reverse plug settings 150 A and 300 A on a $500/5 \text{ CT}$, same dials	No	Eq. (12): $1.25 \times 252.2 \text{ A}$ reverse load current = 315.2 A , then the same rule as item 5.
8	Bolted fault, DG in service; LLL at three-phase buses, LL at two-phase buses (645, 646, 684), LG at single-phase buses (611, 652)	No	Table IV does not state the fault type or impedance of any row. The largest fault the bus can have was taken.
9	Largest phase current through the recloser is the operating current	No	An overcurrent relay trips on its fastest (highest-current) phase element.
10	Source: study case with the $115/4.16 \text{ kV}$, 5 MVA substation transformer	No	It reproduces the IEEE 13-node short-circuit benchmark within about 2 %. The paper's source impedance is not given.
11	R2's direction is decided by fault location (upstream of R2 = reverse)	Implied	PowerFactory's CDG34 has no directional element. The paper itself uses R2 reverse for the fault at 646.
12	Fuses are Gould-Shawmut A055C (E-rated); t_{MMT} and t_{TCT} are read from the library curves, log-log between points	No	The paper gives no fuse make and no fuse curve data, only eq. (6) with $a = -1.8$. A real curve is needed to get a time; PowerFactory's own fuse model returns the same times (item 4).
13	Fuse sizes: F-DL 400E, F632 500E, F633 400E, F646 400E, F652 150E, F671-1 400E, F671-2 300E, F684 200E, F692-R 250E	Partly	Starting sizes from the paper's figures and ref. [8]; then raised by step 9 of the method until the fuse outlasts the fast trips with the DG in service. The sizes behind the paper's Table IV are not printed.
14	The fuse time is taken at the current through the fuse (largest phase), which includes the DG share	Implied	The fuse melts on its own current. Paper, Fig. 15: the fuse F646 carries 3.64 kA while R2 carries 1.54 kA .

7. Why the values differ from the paper's Table IV

R1. The equation and settings reproduce the paper's worked example, so the difference is in the current fed into the equation. Example: at bus 633 the model has 3970 A through R1 and gets 2.111 s; the paper's 1.880 s needs 4287 A. The model's fault levels follow the IEEE benchmark and are lower than the paper's (feeder-head maximum 4.73 kA against the paper's 5.41 kA in Table II). Bus 652 is the opposite: the paper's 13.525 s corresponds to only 1500 A, which is less than the model's LG fault current there, so that row was probably computed for a fault through impedance or a different fault type.

R2. Besides the current, the settings themselves are assumed (items 5 to 7), because the paper does not print them and they cannot be read back from Table IV. The fast times come out between 37 % below and 15 % above the paper's; the delayed times between 30 % below and 183 % above. These R2 values should be read as the result of the stated method, not as a match to the paper.

Fuses. The fuse times differ from the paper's by -65 % (bus 675) to +200 % (bus 692). A fuse curve is steep (t falls roughly with I^{-2} to I^{-3}), so a 20 % difference in current already changes t_{MMT} by 40 to 70 %, and a different fuse size by more. The paper gives neither the sizes nor the fuse currents behind its column, so this column is the result of the method on this model; what can be checked is the criterion itself: t_{MMT} is longer than the last fast trip in 9 of 9 rows (margin column above).